

quality of training data. Ideally, users should give clear prompts with additional contextual clues that guide the model to provide more accurate and evidence-based responses. This is not always possible in the real world, as not every user is capable of or inclined towards giving perfect prompts. So, the system must undergo adversarial training with challenging prompts and inputs, which are bound to induce hallucinations. The model must also be taught to focus on relevant portions of the prompt, gradually eliminating unwanted information or distractions that may trigger wrong outputs.

Step-by-step verification: “Many of the LLM platforms today also have reasoning models, which take the user inputs or prompts and ensure that it is broken down into steps, and the logic of each step is verified before responding,” says Venkatraman.

Feedback mechanisms: Users should be able to flag incorrect information and provide feedback on the relevance of LLM output. The model should ‘learn’ from this feedback to generate more accurate and relevant responses over time. Model developers are also trying to work in some kind of reward or scoring mechanism—wherein the model is rewarded for correct answers and penalised for overconfident but incorrect answers and unfounded extrapolations (hallucinations)—to help it evaluate its own capabilities and stay grounded!

Ensemble methods: Rather than rely on a single model, ensemble methods combine multiple models or ‘learners’ (like regression models or neural networks), leveraging their collective strength to produce more accurate predictions. There are many ensemble methods like bagging, boosting, stacking, and voting, and the right method will depend on the task, data, problem complexity, and available computational resources.

“I always struggle a bit when I’m asked about the ‘hallucination problem’ in LLMs. Because, in some sense, hallucination is all LLMs do. They are dream machines... Only when the dreams go into deemed factually incorrect territory can we label it a ‘hallucination.’ It looks like a bug, but it’s just the LLM doing what it always does... I realise that what people *actually* mean is they don’t want an LLM Assistant (a product like ChatGPT) to hallucinate. An LLM Assistant is a lot more complex system than just the LLM itself, even if one is at the heart of it. There are many ways to mitigate hallucinations in these systems, but using Retrieval Augmented Generation (RAG) to anchor dreams in real data through in-context learning more strongly may be the most common one. Disagreements between multiple samples, reflection, and verification chains. Decoding uncertainty from activations. Tool use. All active and very interesting areas of research.”



— *Excerpts from an X post by Andrej Karpathy, PhD, Founder of Eureka Labs, an AI education initiative; formerly Director of AI at Tesla and Co-founder, OpenAI.*

Stay alert

Indeed, models are improving with better training, and more importantly, the usage of models is also becoming more advanced. More grounding, verification, and safety are being applied. All this reduces hallucinations to a great extent. Hallucination is, after all, an AI model trying to fill a gap between pieces of knowledge that it has learnt and trying to extrapolate. With better training, inference and grounding with web data, these gaps reduce significantly.

Still, in the current scenario, it is very important for people and enterprises using AI to stay alert and

verify the responses, especially when making critical decisions. “Users can detect hallucinations by doing their own research and validating the output. Both users and enterprises can verify output from the LLM systems by doing their own research via Google and other reliable news sources. Users should not rely on information from LLM platforms on domain-specific questions like medical, legal, and investing advice,” suggests Venkatraman.

At the end of the day, remember that AI is not your guru. It is just a tool created by man to ease your life—use it well, with prudence, and all will be well. **END** 🐧

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This article was originally published in the February 2025 issue of Electronics For You.